

AWS EC2 Instance Provisioning using Pulumi

- We will provision the AWS EC2 Instance using Pulumi as an Infrastructure as Code.
 - We will deploy it in a custom Virtual Private Cloud for isolation.
 - We will SSH into the EC2 Instance, and install the docker.
 - Then, we will deploy the Nginx Container and try accessing it on the Web Browser.
-

Prerequisites

1. An AWS account.
 2. AWS CLI installed and configured with the appropriate Azure User or Service Principal.
 3. Pulumi Installed.
-

Write Pulumi Configuration files

First, we will initiate and edit Pulumi configuration files for AWS resources using predefined Pulumi Library available on the internet.

Steps

1. Create a Pulumi Project directory.
2. Open the Powershell.
3. Change the directory to the above-created Pulumi Project.
4. Run the **pulumi new aws-python** command to initialize the *pulumi*.
5. Provide the appropriate values to prompts such as *project-name*, *project-description*, *stack-name*, *toolchain*, *region-name*, etc.
6. This will generate some Pulumi files in this directory.
7. Now we will install predefined Pulumi modules.
8. Activate the **venv** by running **venv\Scripts\activate**.
9. Run **pip install git+https://github.com/inflection-sahil/pulumi.git** to install the modules.
10. Deactivate the **venv** by running **deactivate**.
11. Now open the directory in the preferred IDE.
12. Create *commons* folder
13. Inside the folder create *init.py* file.
14. Import the following in the *init.py* file:
 - from inflection_zone_pulumi.modules.aws.vpc import vpc
 - from inflection_zone_pulumi.modules.aws.ec2 import ec2_instance
15. Click [code](#) for reference.
16. Definition of *init.py* is complete.
17. Now create the *values.py* file in the root folder of the above-created project directory.
18. Define the following values:
 - vpc_properties

- `ec2_properties`
19. Click [code](#) for reference.
 20. The definition of *values.py* is complete.
 21. Now navigate to the *main.py* file present in the root folder of the above-created project directory.
 22. Clear the sample code if present.
 23. Import the following:
 - `from commons import vpc, ec2_instance`
 - `import values`
 24. Define the following objects and pass the values & dependencies as an argument:
 - VPC
 - EC2
 25. Click [code](#) for reference.
 26. Definition of *main.py* is complete.
-

Provisioning the Infrastructure

Now we will provision the infrastructure by applying the above-created configuration files.

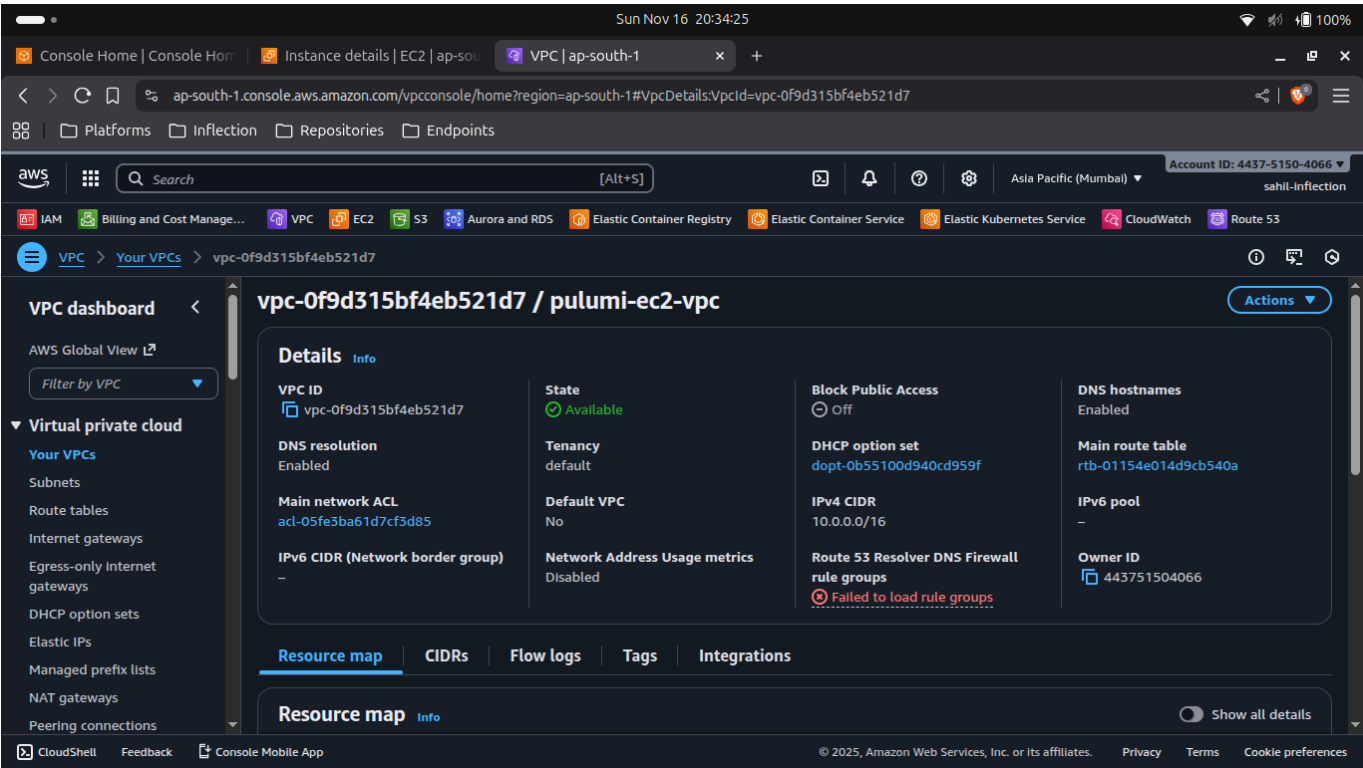
Ensure AWS CLI is configured with the appropriate Credentials.

Steps:

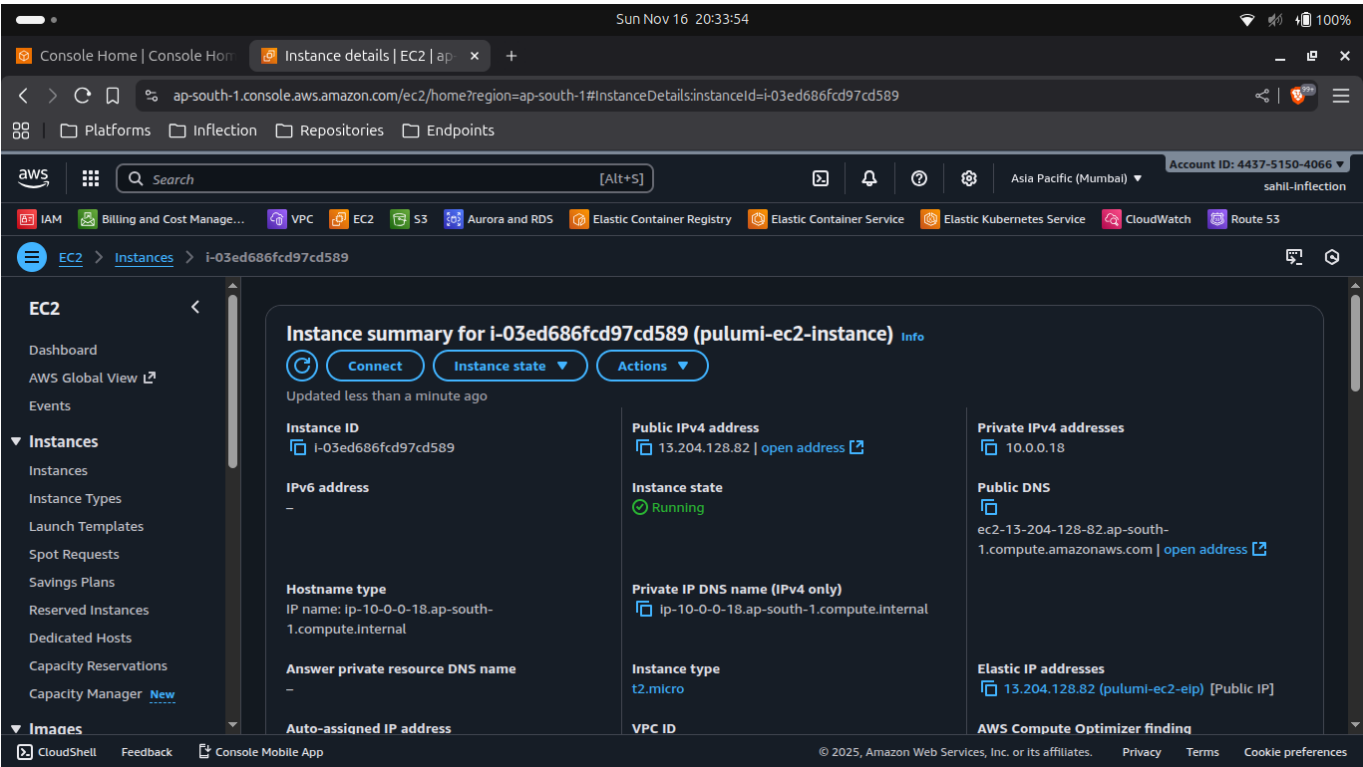
1. Open the PowerShell.
 2. Change the directory to the above-created Pulumi Project.
 3. Run the `pulumi up` command and if prompted, select **yes** to provision the infrastructure onto the Azure Cloud.
 4. Head to the AWS Console, and verify the created resources.
-

Screenshots of Provisioned Infrastructure

VPC Image



EC2 Image



SSH Into EC2 Instance

Now we will SSH into the EC2 Instance and configure it for Nginx container deployment.

Steps

1. Open the Powershell Window.
2. Run the following command to SSH into EC2 Instance and substitute the `<ssh-key-pair>` with the value provided in `values.py` file under `<ec2_properties>` section and `<ec2-public-ip>` with the EC2 Instance Public IP received from **pulumi stack output vm-public-ip** command:

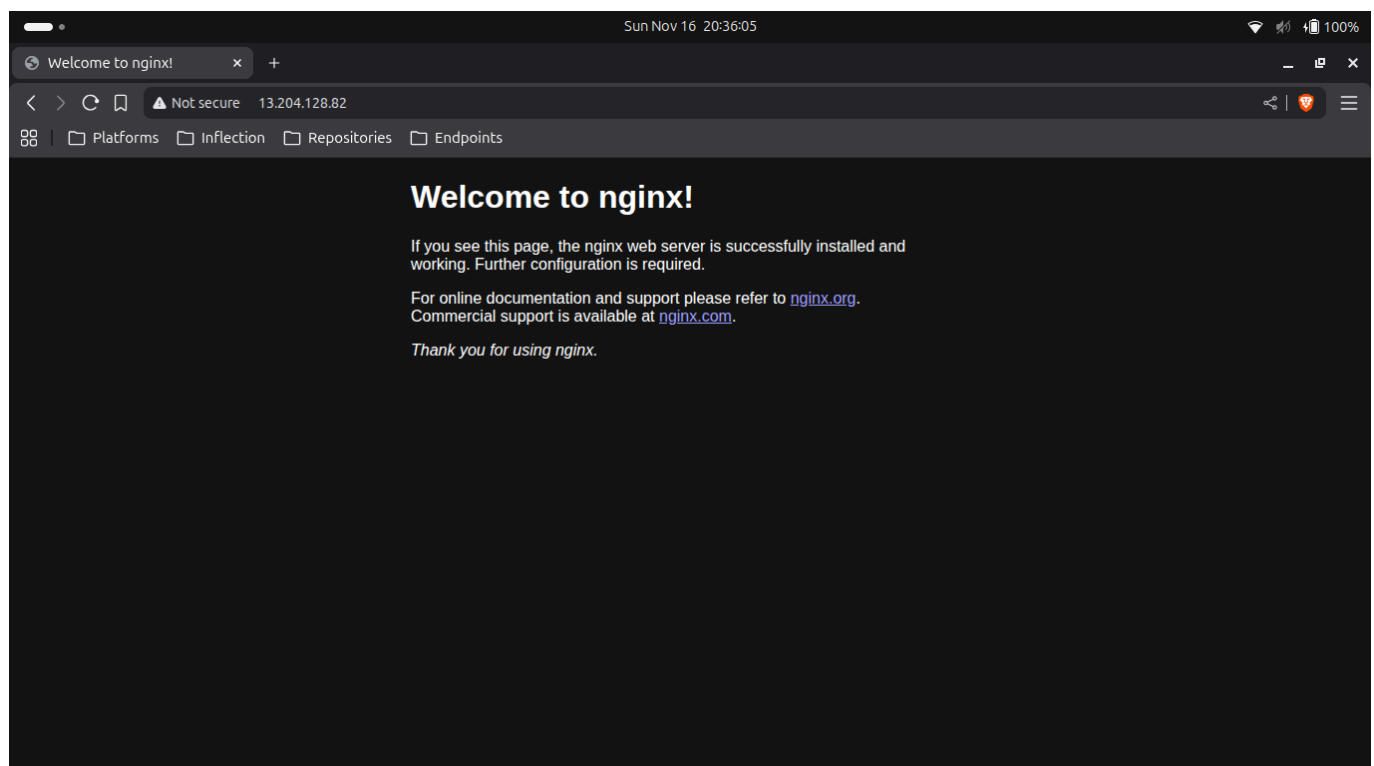
```
ssh -i <ssh-key-pair> ubuntu@<ec2-instance-public-ip>
```

3. Once you enter the server, run the following commands to install the necessary dependencies for deployment and run the nginx container:

```
sudo apt update
sudo apt install -y docker.io
sudo docker run -d -p 80:80 nginx
```

4. Try accessing it on the browser using `<ec2-instance-public-ip>` received from **pulumi stack output vm-public-ip** command.

Nginx Image



Destroy the provisioned infrastructure

Lastly, we will destroy the above-created resources.

Steps

1. To destroy infrastructure, open the Powershell Window and change the directory to the above-created Pulumi Project using the **cd** command.
 2. Run **pulumi destroy** & if prompted, select **yes**.
 3. Infrastructure will be destroyed.
-